

Stage 1: Ingestion

Mobile Artifacts
Target APK / APKS
Multi-Split Packages
Asset Bundles

Statutory Corpus
BB-CSF v1.0 (BRPD)
Draft BD-PDPA
BD-CSA 2023

Stage 2: Extraction

Decompiler Pipeline
APKS Resolver
Dalvik .dex Parser
AXML Decoder

Regulation Engine
7-Tuple Parser
2D Observability Ω
Expert Provenance

Stage 3: Analysis Engine

1. Manifest Security Flags

2. Keystore & Crypto Audit

3. Local Database & Prefs

4. Network & TLS Pinning

5. Third-Party Tracker SDKs

6. Dataflow Taint Propagation

Stage 4: Epistemic Core

Evidence DAG $G=(V,E)$
Statute Nodes
Technical Controls
Bytecode Witnesses



5-State Epistemic Core
States: {S, P, I, O, A}
Observability Gate
Metrics: EC, ESC, SCI

Stage 5: Remediation

Bilingual Reports
Executive Summary
English Diagnostics
Bengali Diagnostics

Developer Tools
Bytecode Offsets
Concrete Code Diffs
CI/CD Gating Rules